

LAB 0

ACTIVE DIRECTORY

Domain Setup • User Creation • Domain Join • Login Verification

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Two client PCs are addressed, pointed at a Windows Server running AD DS, and joined to the sn.lan domain. A single domain user account is then used to log in on both machines, demonstrating centralized authentication.

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Overview & Topology

A Windows Server (Winserver) runs AD DS and hosts the sn.lan domain. Two client PCs — Win2 and Win3 — connect to it through a switch. Each PC is addressed and pointed at the server for DNS, joined to sn.lan, and then logged into using a single domain user account (Asaf1) created once on the server — proving that one set of credentials works across every machine in the domain, which is the whole point of Active Directory.

- Winserver — AD DS + DNS, at 10.0.0.1, domain sn.lan
- Win2 (PC-1) — client, joined to sn.lan
- Win3 (USER-PC) — client, joined to sn.lan
- Asaf1 — single domain user account, used to log into both clients

IP Address Plan

Device	Computer Name	IP Address	Subnet Mask	Preferred DNS
Winserver	SERVER2016	10.0.0.1	255.0.0.0	— (is the DNS server)
Win2	PC-1	10.0.0.10	255.0.0.0	10.0.0.1
Win3	USER-PC	10.0.0.20	255.0.0.0	10.0.0.1

TASK 1 IP Assignment (PC1, PC2, Server)

Static IPs set on both clients and the server so every machine can find the domain controller by address before DNS or AD are involved.

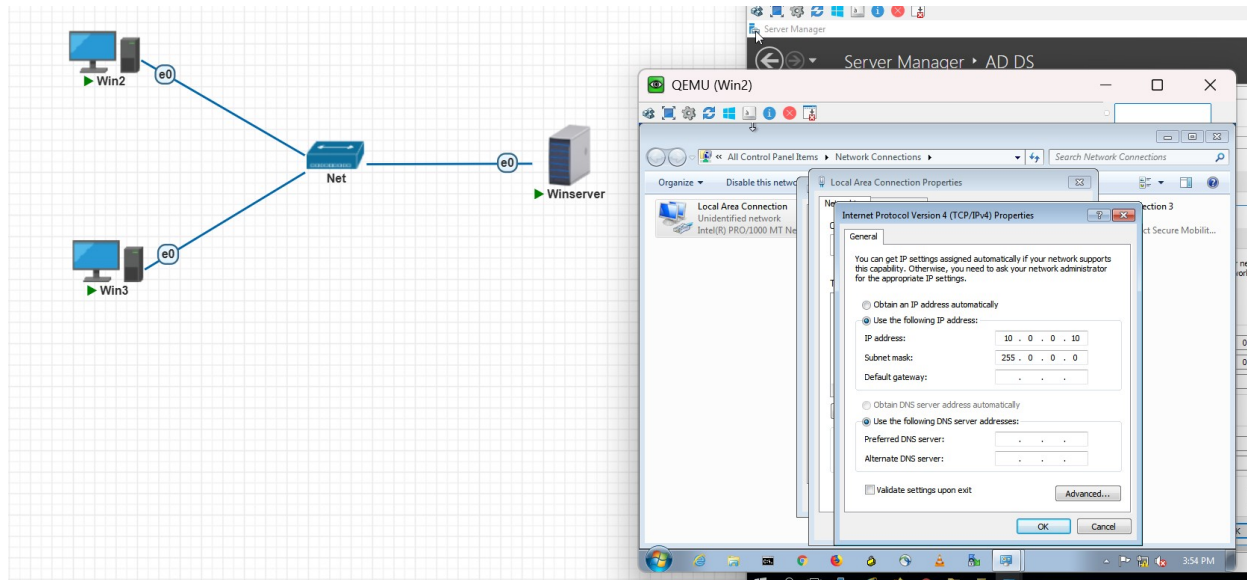


Figure 1 — Topology: Win2 and Win3 behind a switch, connected to Winserver

- Win2 (PC-1): 10.0.0.10 / 255.0.0.0, preferred DNS 10.0.0.1

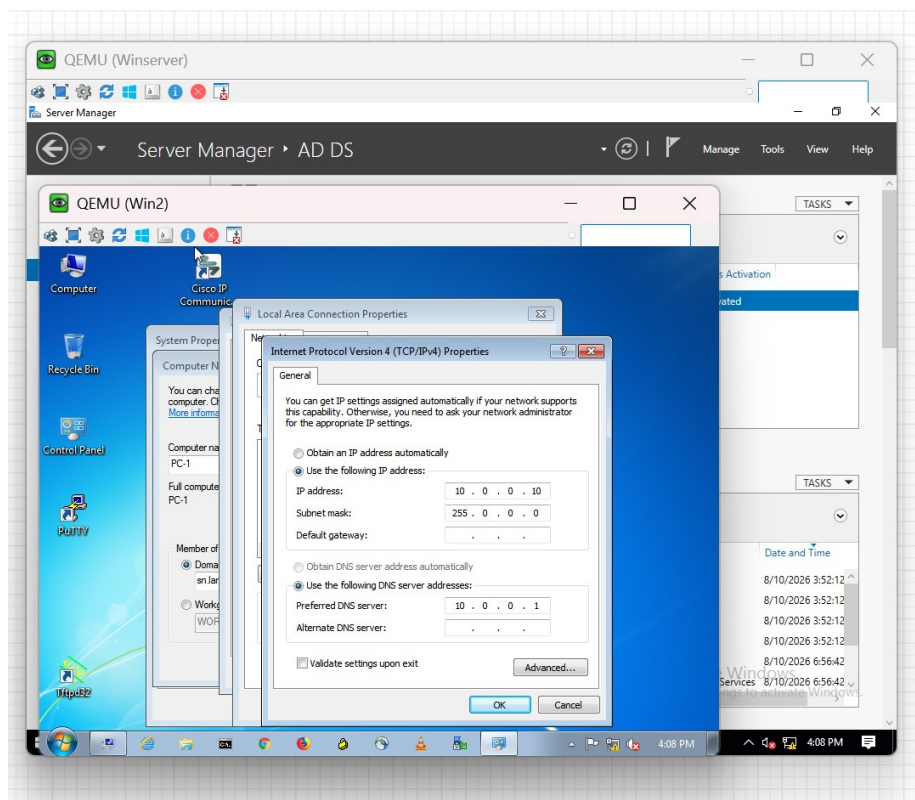


Figure 2 — Win2: IP and DNS configured, computer name PC-1

- Win3: 10.0.0.20 / 255.0.0.0, preferred DNS 10.0.0.1

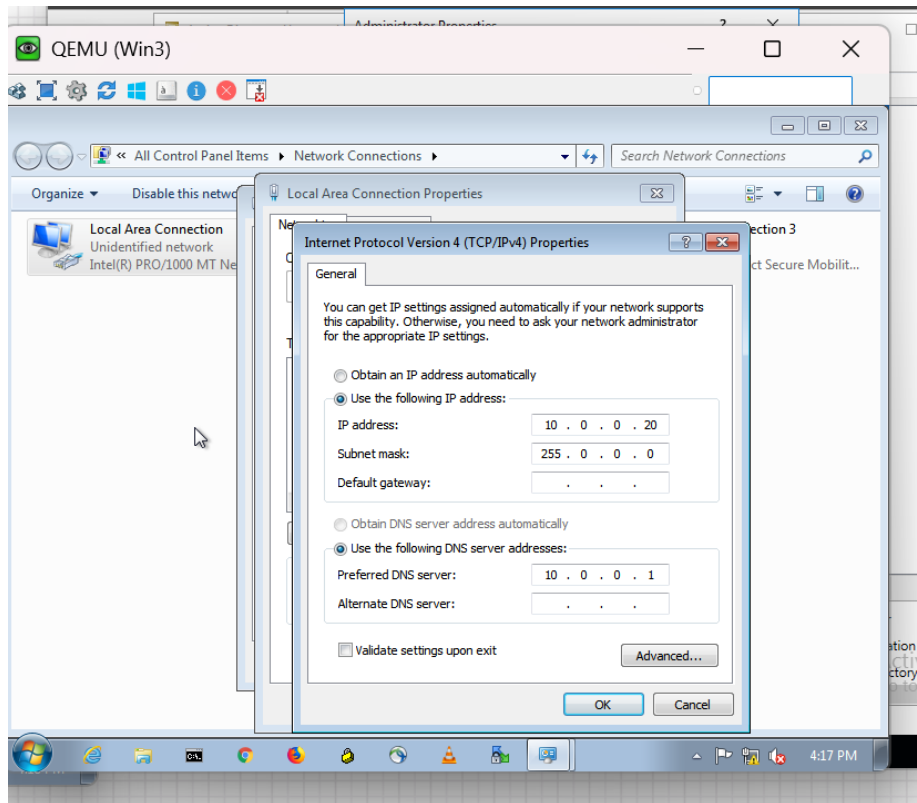


Figure 3 — Win3: IP and DNS configured

- Winserver: 10.0.0.1 / 255.0.0.0 — the domain's own DNS server, so no external DNS entry needed

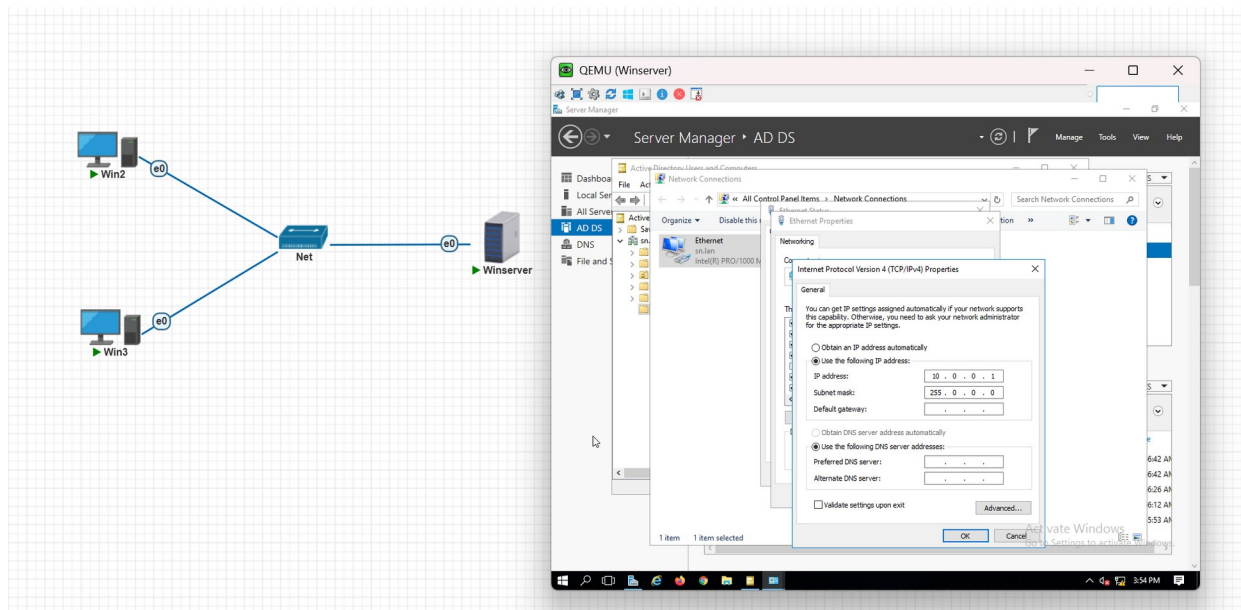


Figure 4 — Winserver: IP configured, topology and AD DS console visible

TASK 2 Creating the Domain User

A new user is created in Active Directory Users and Computers, inside `sn.lan/Users`.

- Full name: Asaf1
- User logon name: Asaf1@sn.lan

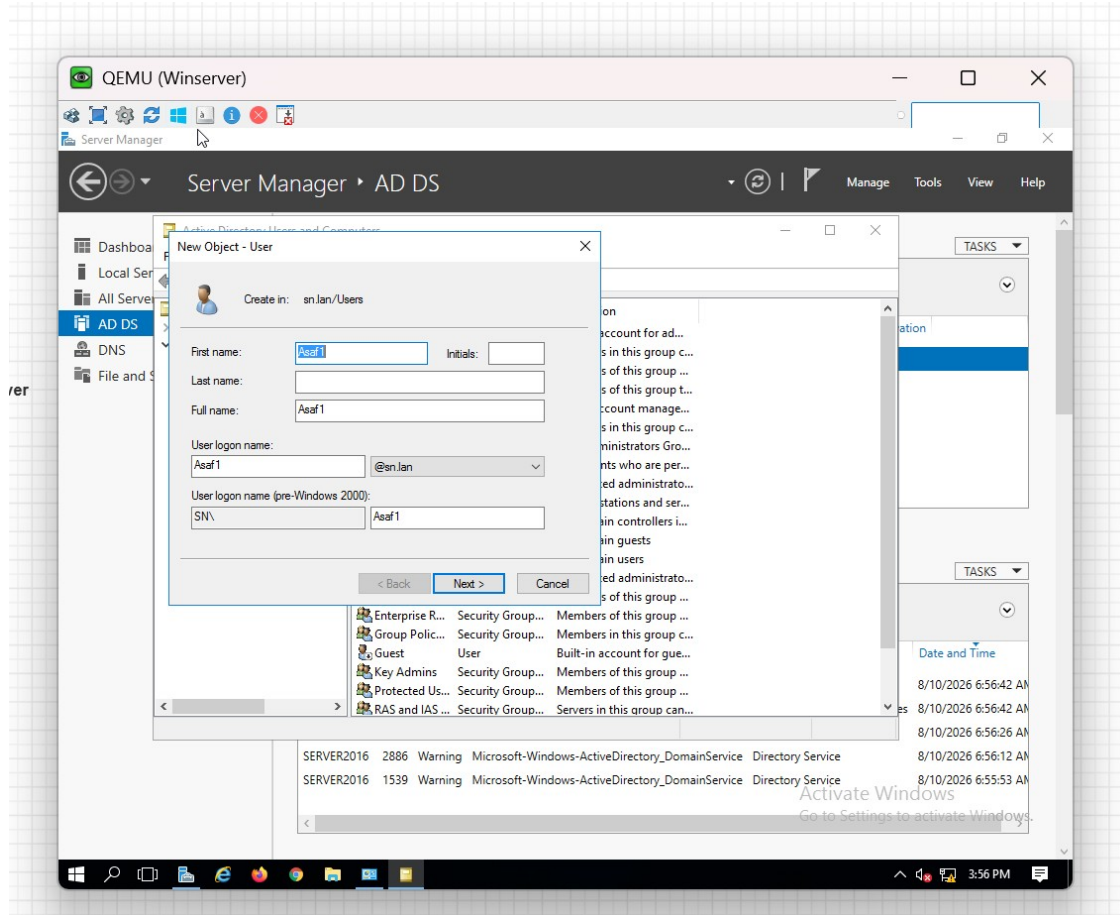


Figure 5 — New Object - User: creating Asaf1 in sn.lan/Users

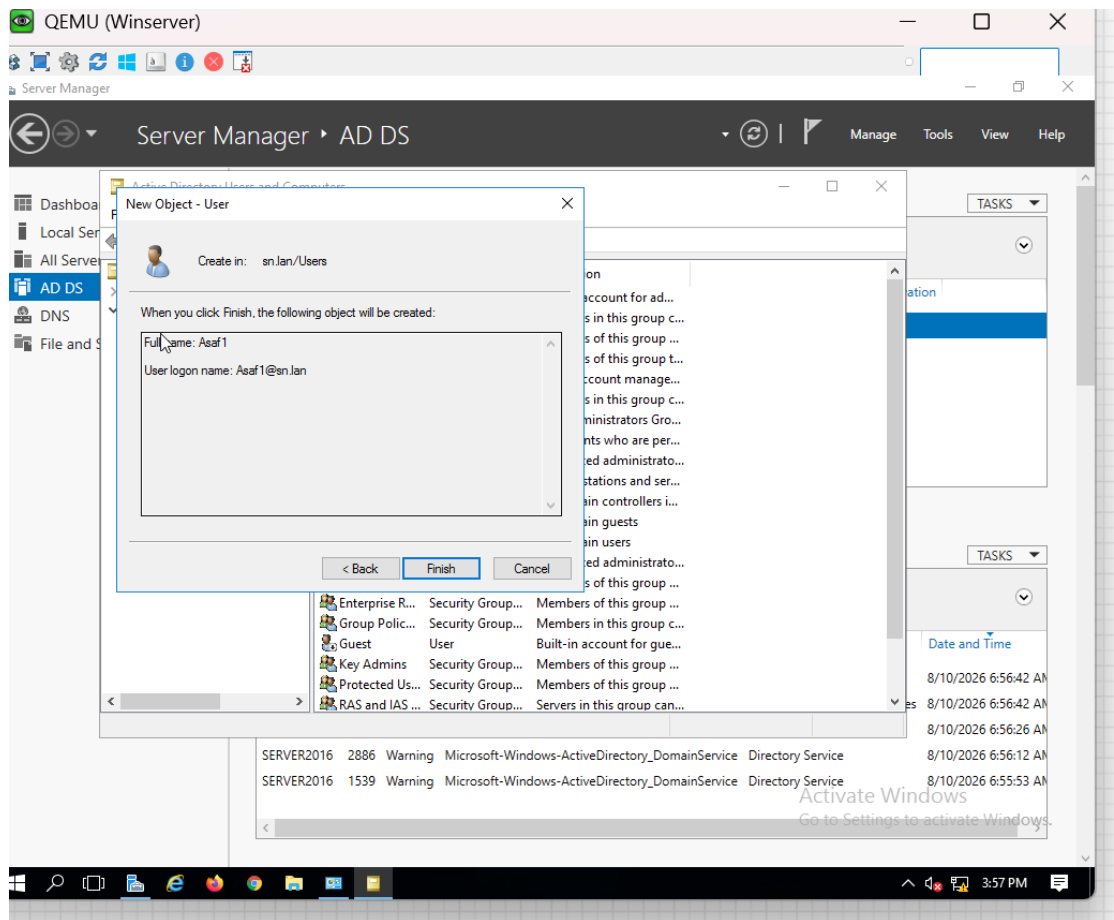


Figure 6 — Confirmation: Asaf1 / Asaf1@sn.lan ready to be created

TASK 3 Joining PCs to the Domain

Each client's System Properties → Computer Name/Domain Changes is pointed at sn.lan and joined using server credentials.

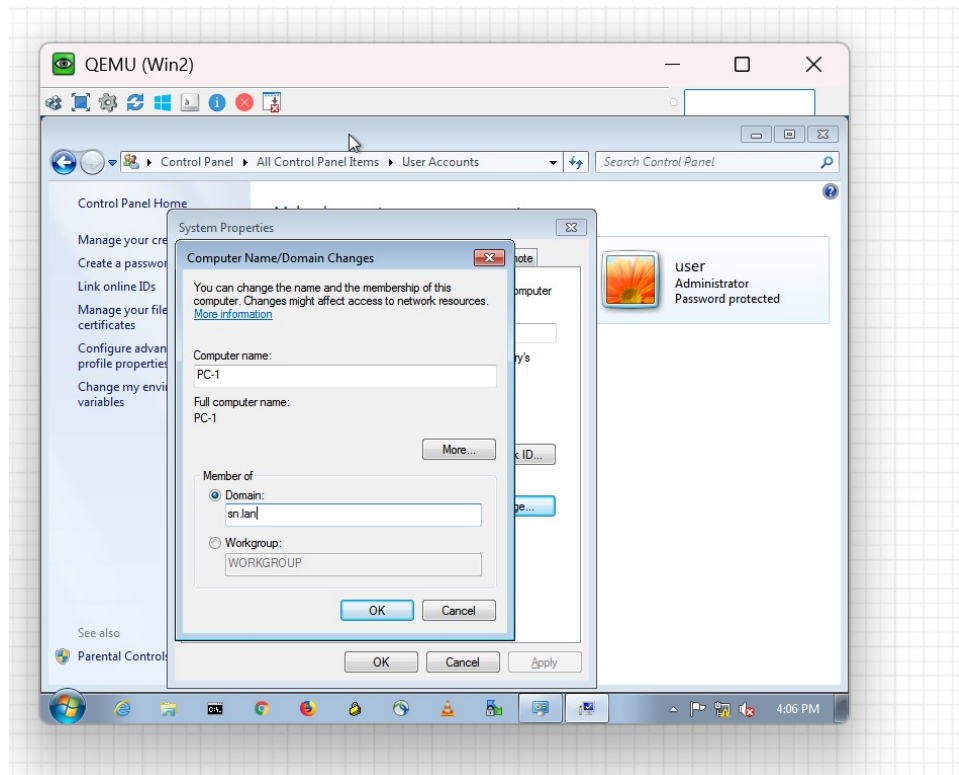


Figure 7 — Win2 (PC-1): joining the sn.lan domain

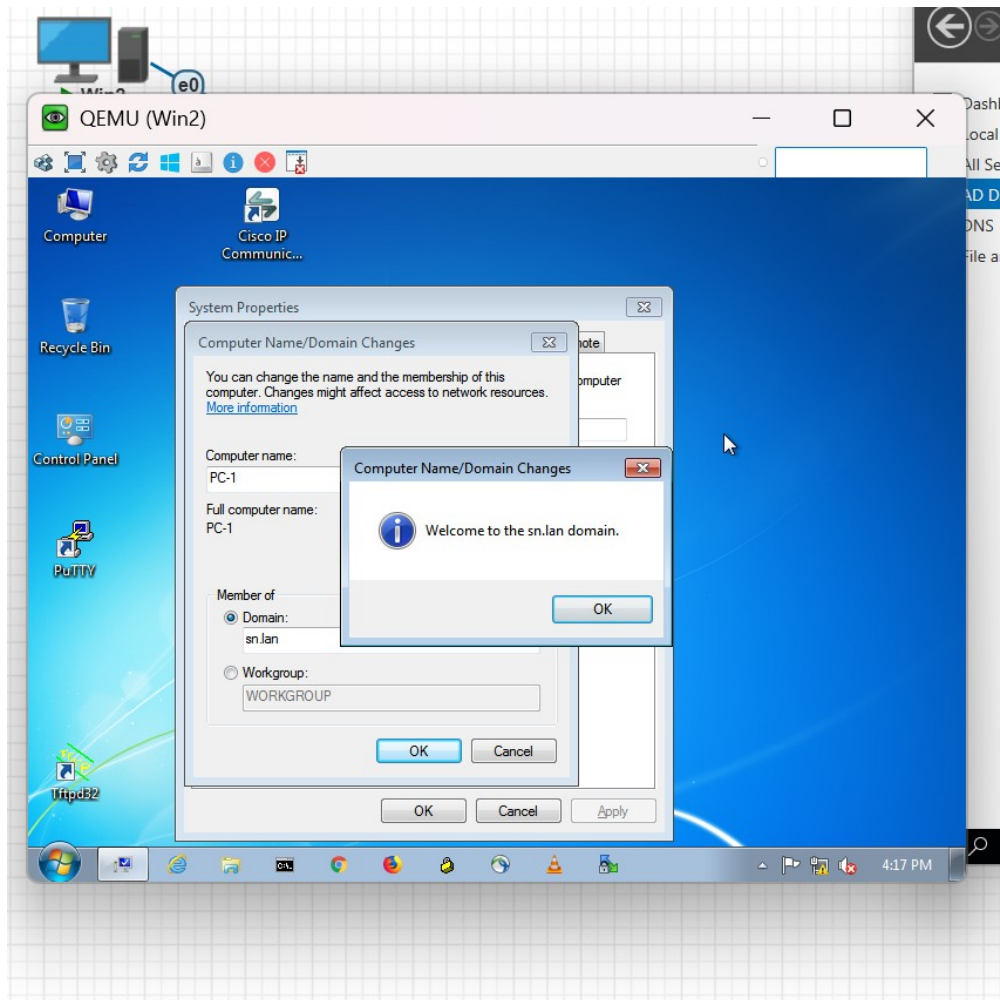


Figure 8 — Win2: “Welcome to the sn.lan domain”

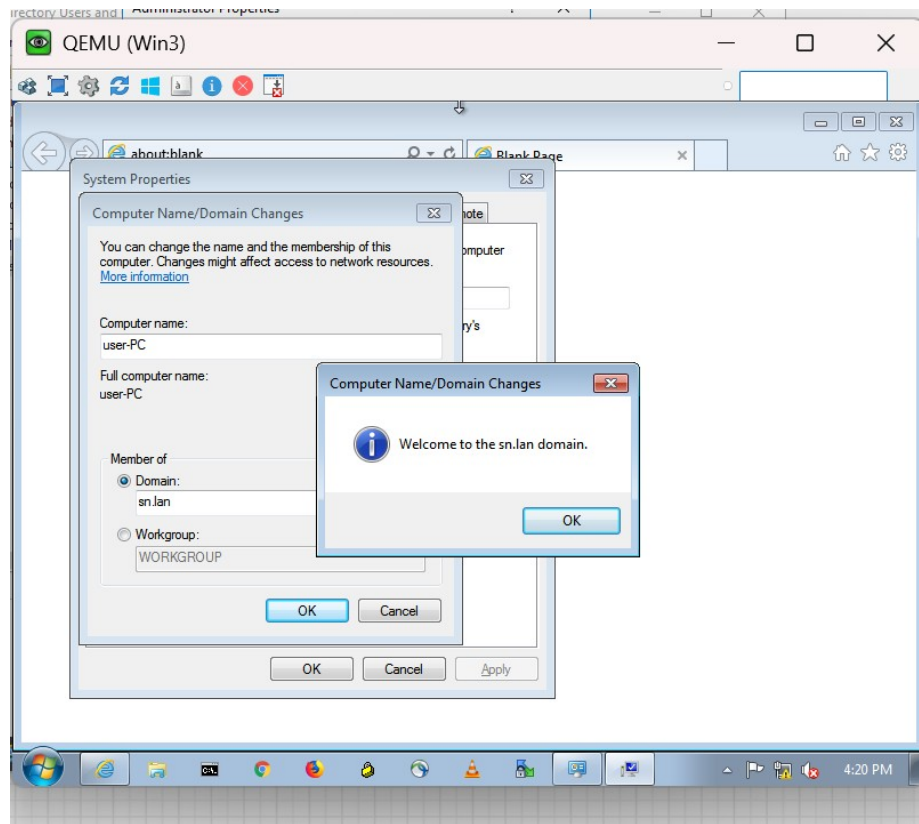


Figure 9 — Win3 (USER-PC): "Welcome to the sn.lan domain"

After a restart, Win2's System Properties confirms the full computer name is now PC-1.sn.lan under domain sn.lan.

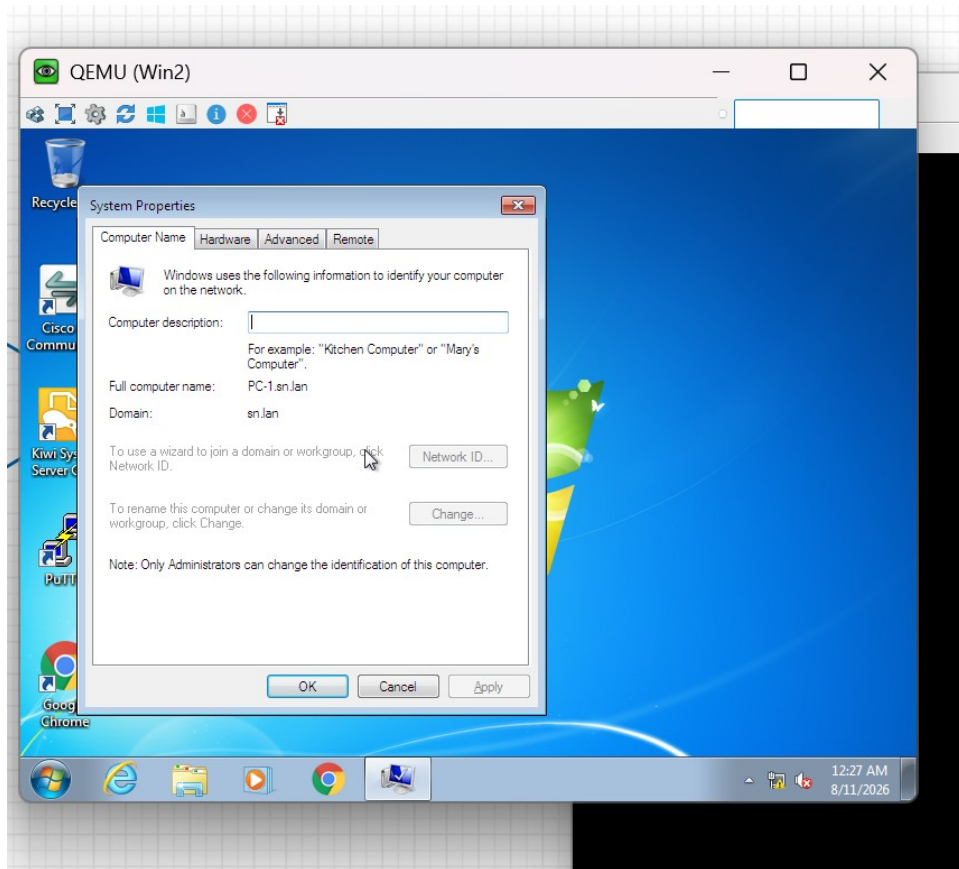


Figure 10 — Win2: full computer name PC-1.sn.lan, domain sn.lan confirmed

TASK 4 Verifying Domain Login

The same domain account, Asaf1, is used to log in on both clients — proof that authentication is centralized on the server rather than per-machine.

- Win2: logs on as sn.lan\asaf1

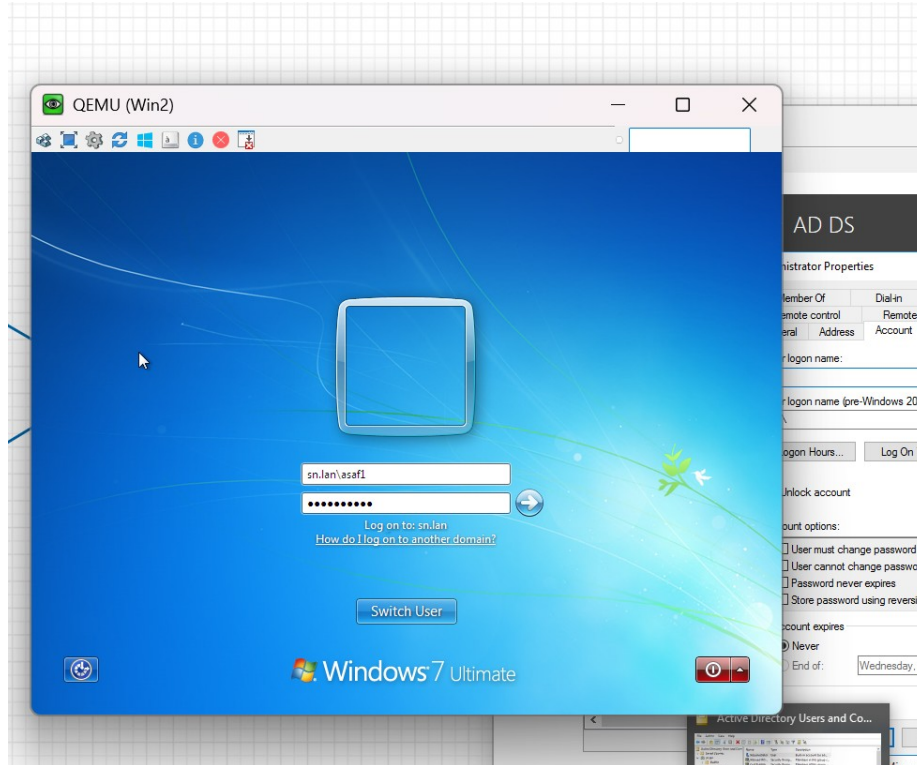


Figure 11 — Win2 logon screen: sn.lan\asaf1

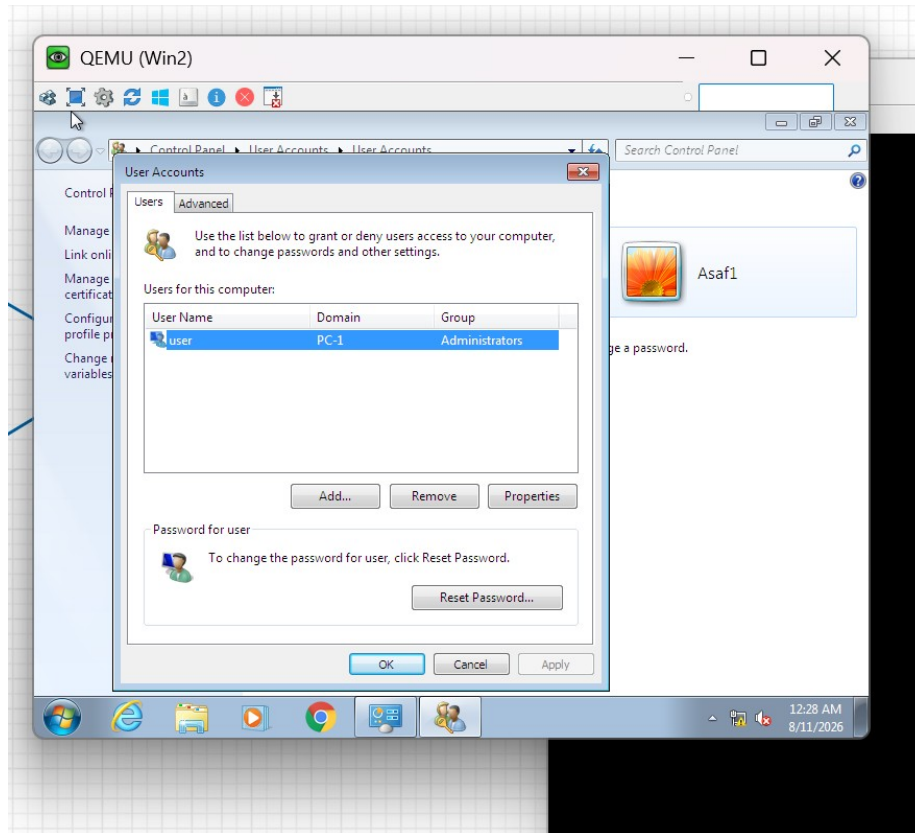


Figure 12 — Win2: signed in as Asaf1 (User Accounts panel, top right)

- Win3: logs on with the same sn.lan\asaf1 account

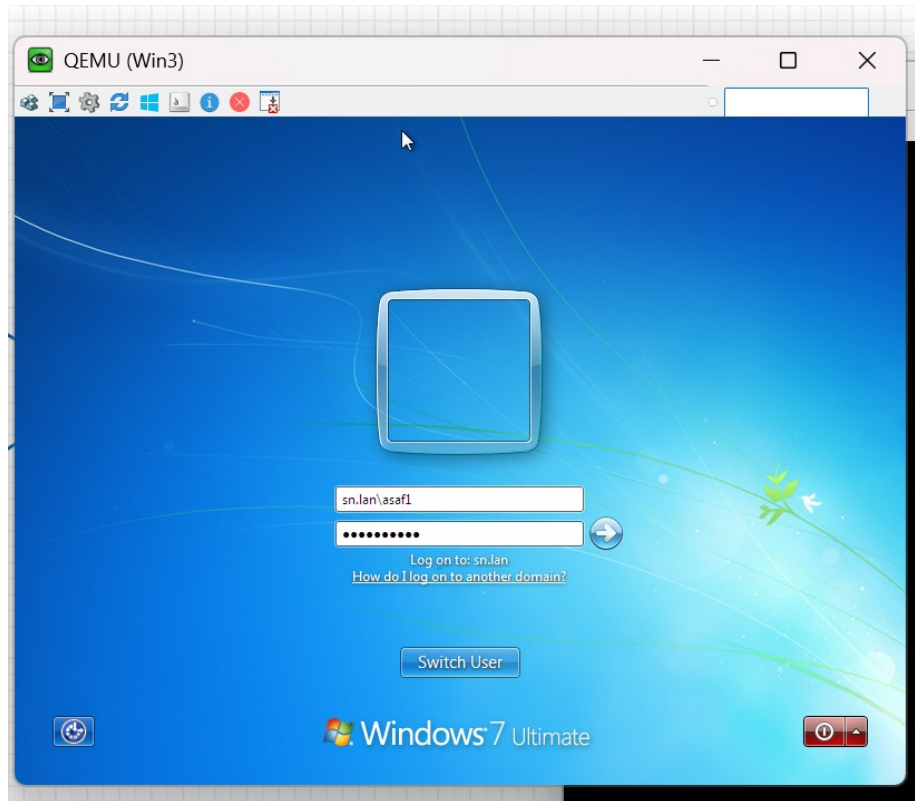


Figure 13 — Win3 logon screen: sn.lan\asaf1

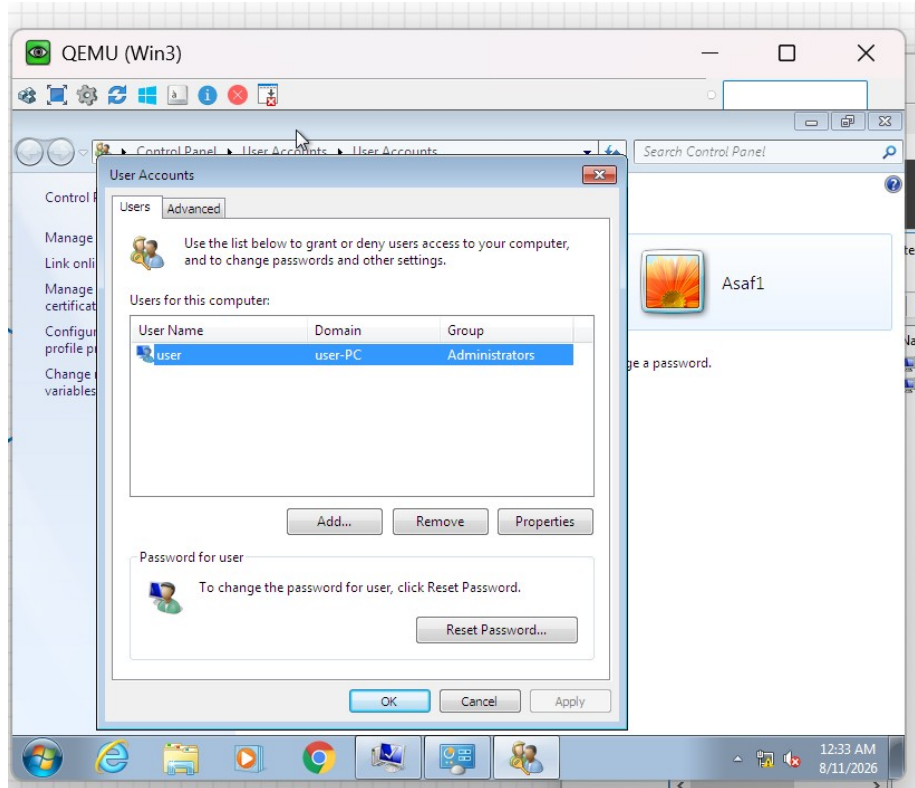


Figure 14 — Win3: signed in as Asaf1 (User Accounts panel, top right)

TASK 5 Verifying Computer Objects in Active Directory

Back on the server, both machines now appear as Computer objects under `sn.lan` — confirming the domain join succeeded for each.

- PC-1 — Computer
- USER-PC — Computer

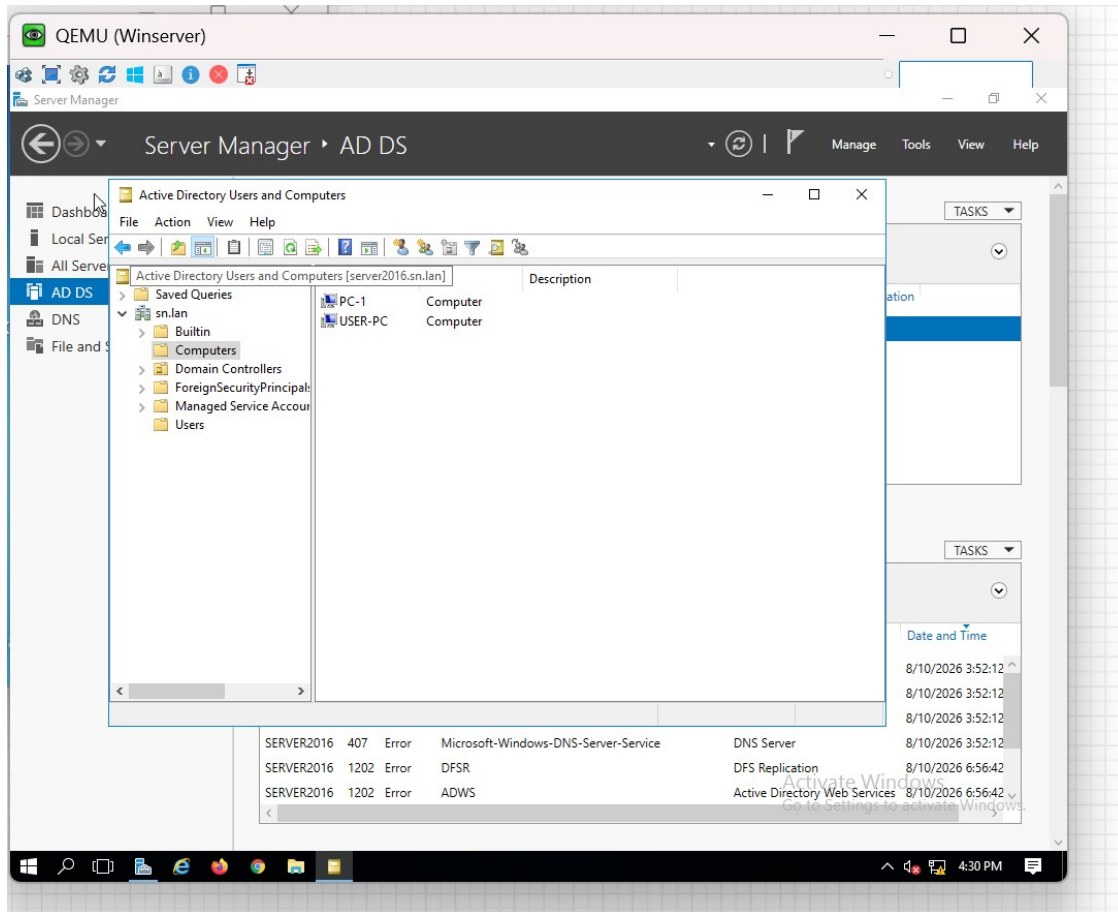


Figure 15 — Active Directory Users and Computers: PC-1 and USER-PC both listed under sn.lan

Conclusion

Both client PCs were addressed, joined to the sn.lan domain, and confirmed as Computer objects in Active Directory Users and Computers. A single domain user, Asaf1, created once on the server, successfully logged into both Win2 and Win3 — demonstrating the core benefit of Active Directory: one identity, managed centrally, usable across every machine in the domain.